

Kyaw Soe Lwin

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SUMMARY

Data Science student and aspiring AI/ML Engineer with hands-on experience building Python-based machine learning pipelines, computer vision applications, and Generative AI/LLM projects. Proficient in Pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow, PyTorch, and Hugging Face Transformers, with practical exposure to data preprocessing, feature engineering, model evaluation, RAG, and cloud deployment (AWS, GCP, Vertex AI). Strong analytical foundation from a 4.0 GPA in Data Science coursework, plus completed specializations in Generative AI, Deep Learning, and Machine Learning. Quick learner eager to grow into production-level AI/ML development.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Java, Bash, Pandas, NumPy, Data Preprocessing, Feature Engineering, Data Manipulation

Machine Learning: Scikit-Learn, XGBoost, Model Evaluation, Supervised/Unsupervised Learning, Optuna (Hyperparameter Tuning), SHAP (Explainability)

Deep Learning, NLP & Generative AI: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CNNs, Computer Vision, Instruction Fine-Tuning (PEFT/LoRA), RLHF, LLM Evaluation, Prompt Engineering, RAG, Vector Databases (Chroma), LLM APIs (Anthropic Claude)

Cloud & DevOps: Amazon Web Services (AWS), Google Cloud Platform (GCP), Vertex AI, Docker, Kubernetes (GKE), Streamlit, Git/GitHub, Linux

Software Methodologies: Agile/Scrum Development, Automated Unit Testing, Modular Pipeline Architecture, Object-Oriented Programming (OOP)

EDUCATION

University of California, San Diego

Bachelor of Science in Data Science (Incoming Transfer)

San Diego, CA

Expected June 2028

Skyline College

Associate Studies in Data Science | GPA: 4.0

San Bruno, CA

Aug 2024 - May 2026

• **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (Java), Linear Algebra, Multivariable Calculus

TECHNICAL PROJECTS

Local Expert - PDF QA Engine (RAG Pipeline) | Python, Hugging Face Transformers, Chroma, Anthropic API, Streamlit 2026

- Architected an end-to-end Retrieval-Augmented Generation (RAG) system in Python for grounded question-answering over private PDFs, hand-building every stage - chunking, embedding, vector indexing, retrieval, and prompt assembly - for full control and inspectability over retrieval quality and citation accuracy.
- Built a document ingestion pipeline that parses PDFs page-by-page into overlapping ~800-character chunks, embeds them locally via Hugging Face `sentence-transformers`, and indexes them in a local Chroma vector database for cosine-similarity retrieval.
- Designed a swappable LLM provider interface (Anthropic Claude API and local Ollama) behind a shared configuration layer, delivering answers grounded in retrieved context with citations back to source file and page through a Streamlit chat UI and CLI.

Telco Customer Churn Prediction Pipeline | Python, Scikit-Learn, XGBoost, Optuna, SHAP, Streamlit 2026

- Developed an end-to-end, leakage-free machine learning pipeline in Python on 7,043 structured customer records, applying `ColumnTransformer`-based feature preprocessing on a stratified 70/15/15 train/validation/test split to ensure reliable model evaluation.
- Optimized model performance through automated hyperparameter tuning (30-trial Optuna Bayesian search) to improve predictive accuracy, achieving 0.844 ROC-AUC / 0.671 PR-AUC with a <3.5 point generalization gap verified across leakage audits.
- Built and deployed a Python web application (multi-tab Streamlit dashboard) integrating live what-if simulation, SHAP model explainability, and cost-sensitive ROI modeling, turning model predictions into actionable retention decisions (\$20 cost / \$200 LTV).

Real-Time Facial Emotion Detection Engine | Python, OpenCV, TensorFlow, Keras, mini-XCEPTION 2026

- Built a real-time computer vision application using TensorFlow and Keras, training a lightweight mini-XCEPTION CNN with depthwise separable convolutions on the FER-2013 dataset to classify facial emotion from live video.
- Engineered a low-latency, multi-frame OpenCV inference pipeline in Python that integrates model predictions into a live UI, rendering probability distributions across 7 discrete emotion classes in real time.

Priceout Collective - Affordability Policy Simulator | Python, Pandas, XGBoost, Scikit-Learn, JavaScript 2026

- Built a Python data pipeline (Pandas, Scikit-Learn) at the Building for Good Hackathon, preprocessing and modeling 1.17M household records across 4 policy variables to precompute 945 scenario combinations for zero-latency simulation.
- Diagnosed and eliminated target leakage in an XGBoost classification model through rigorous feature evaluation, improving model reliability to 97.3% accuracy / 0.998 AUC and validating results against ALICE and HUD CHAS housing benchmarks.

CERTIFICATIONS & SPECIALIZED TRAINING

DeepLearning.AI, Stanford Online & AWS

Advanced AI, Large Language Models, and Mathematical Foundations

Coursera

2025 - 2026

- Generative AI with Large Language Models - DeepLearning.AI & AWS
- Deep Learning Specialization - CNNs, Sequence Models, Hyperparameter Optimization
- Machine Learning Specialization - Supervised/Unsupervised Learning, Recommenders, RL
- Mathematics for Machine Learning and Data Science - Linear Algebra, Calculus, Probability & Statistics

Google Cloud, IBM & University Programs

Applied Generative AI, Cloud Systems, and Software Engineering

Industry Credentials

2025 - 2026

- Google Cloud Skills Boost Portfolio - Vertex AI, LLM Prompting, Model Tuning, Chaiyo GCP (Seasons 4 & 6)
- Frontier Tech Leaders Programme - Myanmar Machine Learning Bootcamp, UNDP
- Applied Python & Agile Scrum - Harvard (CS50P), Univ. of Michigan & IBM

EXPERIENCE & HONORS

Teacher Ni Language Centre

Academic Tutor

International

Oct 2022 - Dec 2023

- Mentored cohorts of 40+ students through structured technical curricula, conducting weekly evaluations to track learning outcomes and provide targeted feedback.

Jack Kent Cooke Undergraduate Transfer Scholarship | Semifinalist

2026

Sterling Redman Scholarship & F.L. Griffin Scholarship | Skyline College Recipient

2025 - 2026